DSA 3050A: ADVANCED POWER BI EXAMINATION
End-Semester Practical Examination Report

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SECTION 1: INTRODUCTION

This project presents an advanced Business Intelligence solution developed using Power BI for stock market performance analysis within the Banking and Finance domain. The objective is to support strategic decision-making by analyzing historical stock prices, sector performance, volatility patterns, and company-level trends across the S&P 500 market.

The solution follows the complete BI lifecycle, including data acquisition, transformation, modelling, DAX development, dashboard design, analytical interpretation, and professional deployment.

The dashboard was designed to provide both executive-level reporting and detailed analytical exploration, allowing decision-makers to identify market trends, investment opportunities, and performance risks across different sectors and companies.

SECTION 2: DATA ACQUISITION AND UNDERSTANDING

The dataset used in this project was sourced from Kaggle and contains real-world S&P 500 stock market data, including daily stock prices and company metadata.

The main transactional table contains approximately 660,000 rows, making it highly suitable for advanced analytical modelling, enterprise-level reporting, and Business Intelligence implementation.

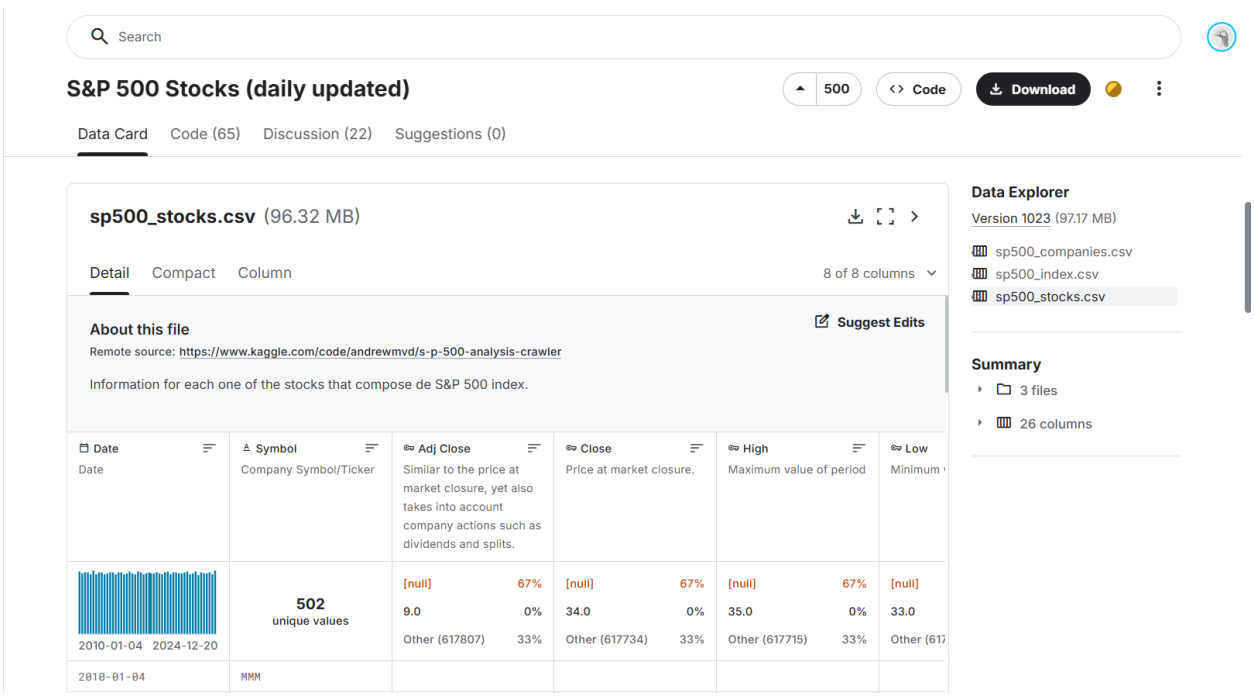
The dataset includes multiple related tables and contains the required numerical fields, categorical attributes, date fields, company identifiers, sector classifications, and geographical information. This supports time intelligence analysis, trend evaluation, and comparative reporting.

The dataset was selected because it offers strong analytical storytelling opportunities and aligns well with the Banking and Finance domain required for this project.

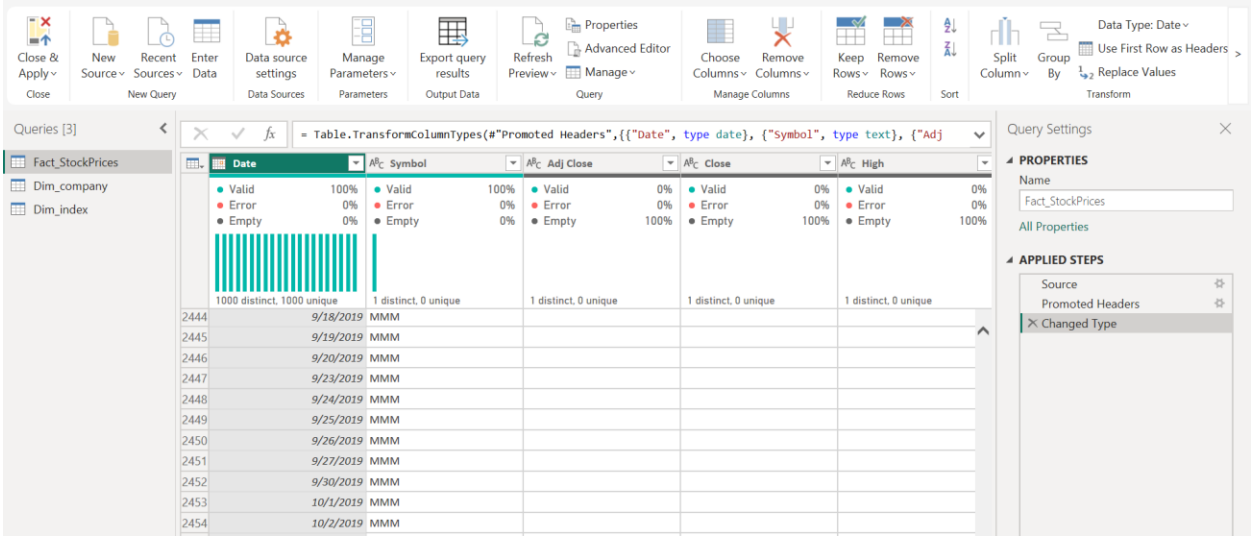
Tables used:
Fact_StockPrices

Dim_Company
Dim_Date

Key fields:
Date
Open
Close
High
Low
Volume
Sector
Company Name
Country
Market Cap



[Figure 2: Dataset source from Kaggle]



[Figure 3: Raw data preview in Power BI]

SECTION 3: DATA CLEANING AND TRANSFORMATION

Extensive data cleaning and transformation were performed using Power Query to ensure data quality, consistency, and analytical readiness.

Major transformation steps included removing duplicates, handling missing values, correcting data types, standardizing inconsistent text fields, removing unnecessary columns, creating derived analytical columns, extracting Year, Month, and Quarter, and building a professional Date dimension table.

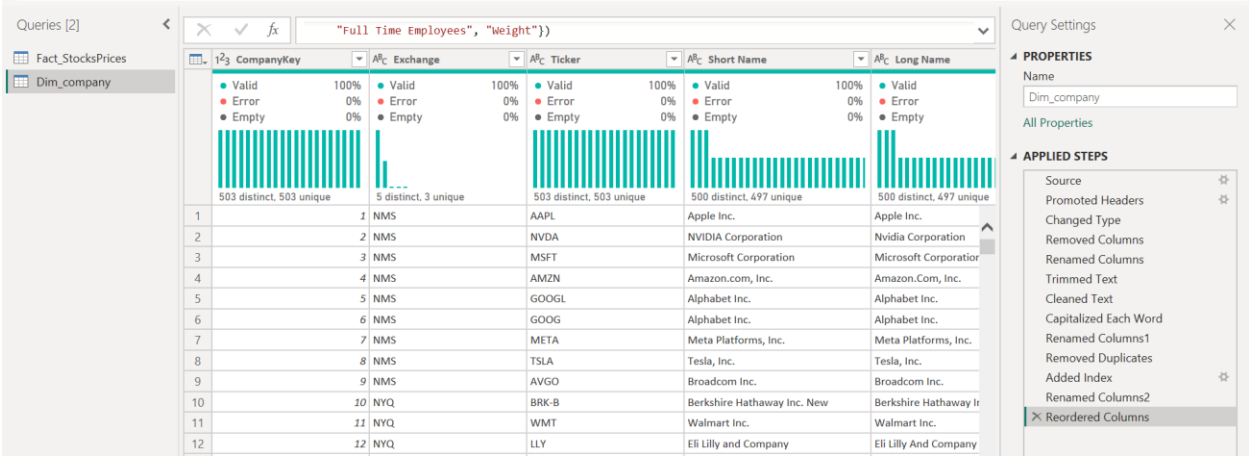
Missing values in key fields such as Close and Volume were addressed appropriately to prevent reporting inaccuracies. Duplicate handling ensured data integrity, while unnecessary columns were removed to improve model performance.

Two important calculated columns were created:

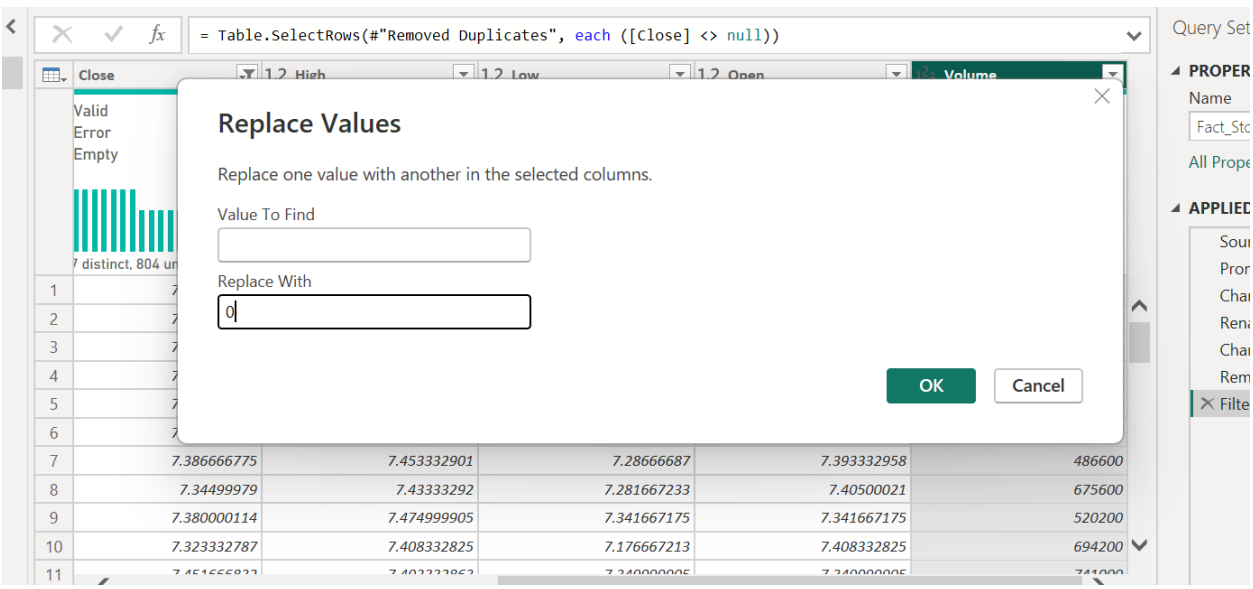
Daily Return %
This supports stock return analysis and volatility measurement.

Price Range
This supports daily trading spread analysis using High minus Low.

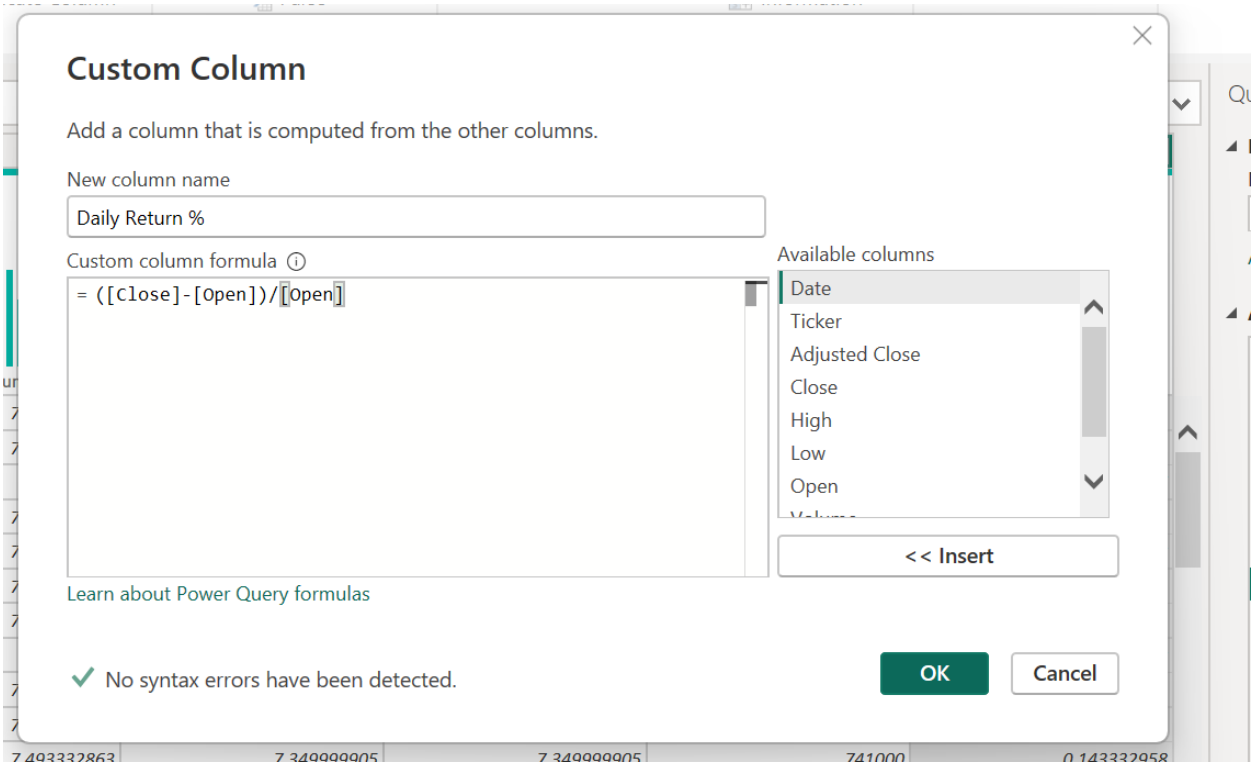
These transformations significantly improved the quality and reliability of downstream reporting.



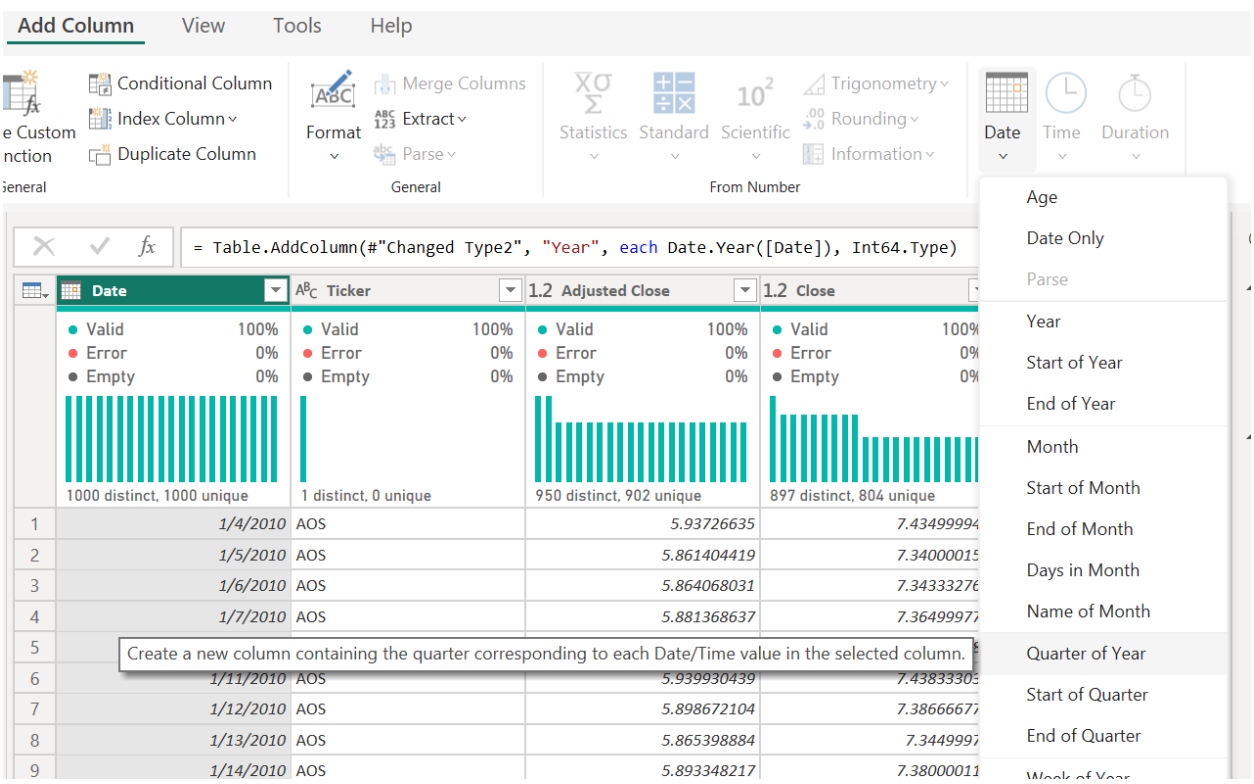
[Figure 4: Applied Steps in Power Query]



[Figure 5: Replacing null values]



[Figure 6: Daily Return % custom column formula]



[Figure 7: Year, Month, and Quarter extraction]

SECTION 4: DATA MODELLING

A clean and efficient Star Schema was implemented to support analytical reporting and improve model performance.

The structure consists of:

Central Fact Table:
Fact_StockPrices

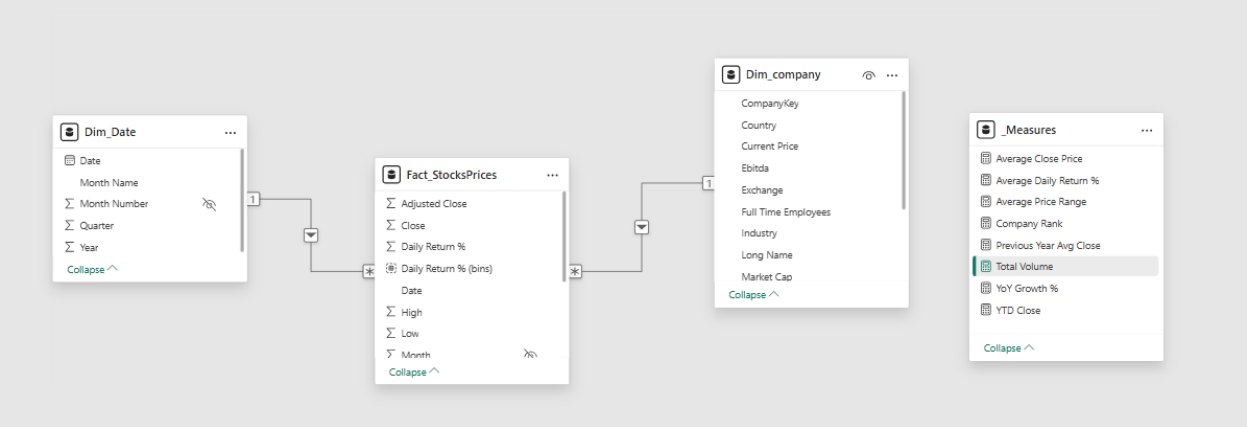
Connected Dimension Tables:
Dim_Company
Dim_Date

Relationships were configured using one-to-many cardinality and single-direction filtering. Hidden technical keys were used to make the report cleaner and easier to use.

This modelling approach avoids many-to-many relationships and ensures accurate aggregations, improved DAX performance, and scalable report development.

The Date table supports time intelligence calculations such as Year-over-Year growth and Year-To-Date performance analysis.

This model aligns with standard Business Intelligence best practices used in enterprise reporting environments.



[Figure 8: Final star schema model view]
[Figure 9: Relationship configuration window]

SECTION 5: DAX MEASURES AND CALCULATED COLUMNS

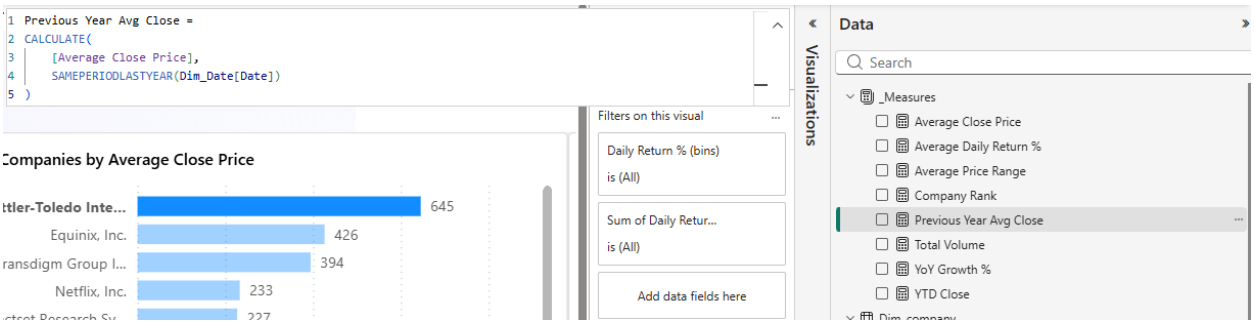
Advanced DAX measures were created to support executive reporting, comparative analysis, ranking, and time intelligence calculations.

These measures move the project beyond descriptive reporting into advanced analytical insight generation.

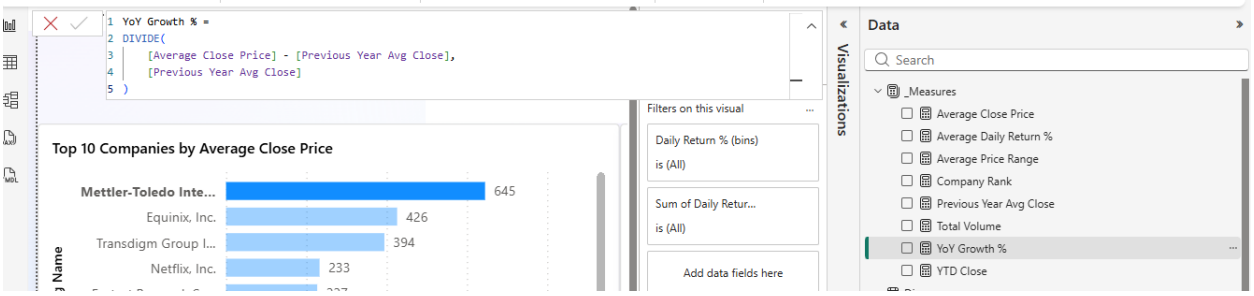
Measures created:
Total Volume
Average Close Price
Average Daily Return %
Average Price Range
Previous Year Average Close
Year-over-Year (YoY) Growth %
Year-To-Date (YTD) Close
Company Rank

Calculated columns created:
Daily Return %
Price Range

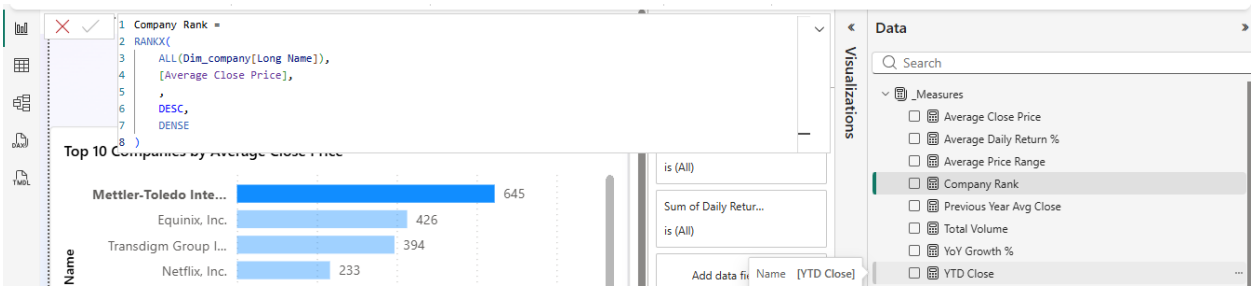
These calculations support growth analysis, volatility analysis, ranking of companies, comparative reporting, and time-series intelligence.



[Figure 10: DAX formula for Previous Year Average Close]



[Figure 11: DAX formula for YoY Growth %]



[Figure 12: DAX formula for Company Rank]

SECTION 6: DASHBOARD DESIGN AND VISUALIZATION

A professional three-page interactive dashboard was developed to support both executive overview and detailed analytical exploration.

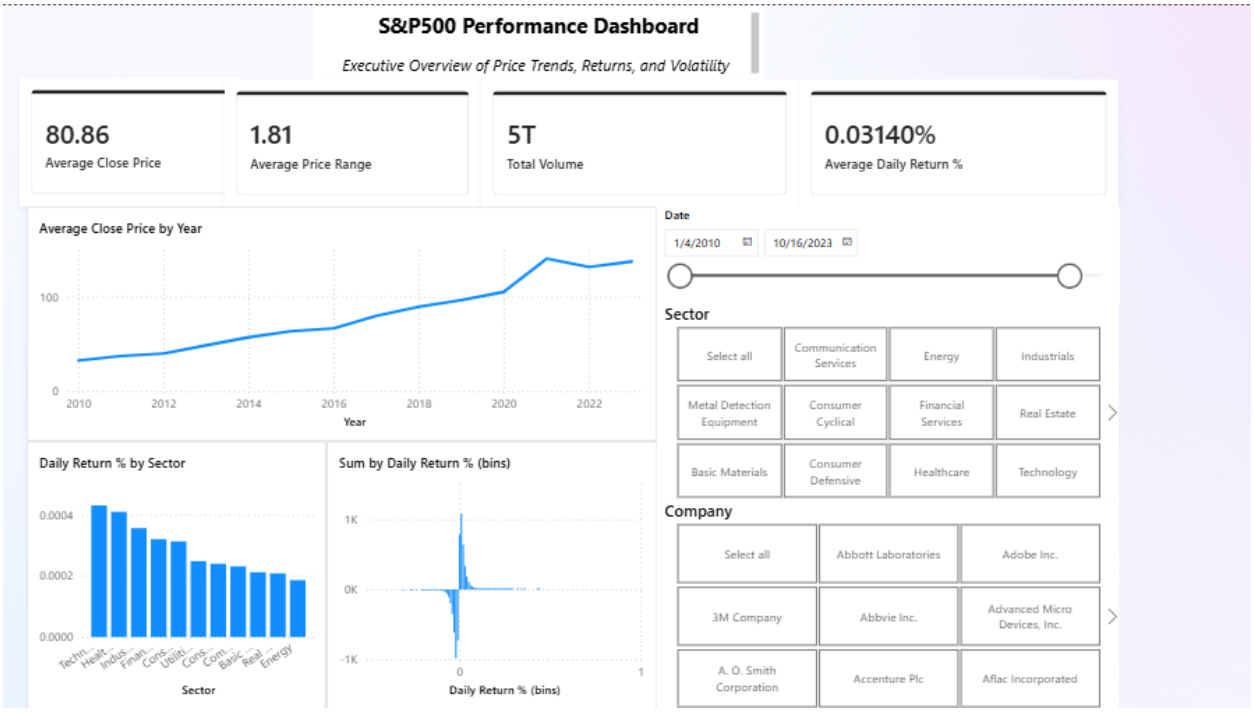
The design focused on clarity, usability, professional layout, and decision-support functionality.

Interactive slicers, drill-down functionality, and cross-filtering improve usability and allow dynamic analysis across sectors, companies, and time periods.

Page 1 — Executive Summary

This page provides high-level KPI reporting and strategic overview using KPI cards, a price trend line chart, sector comparison, market volatility distribution, and interactive slicers.

This supports executive decision-making.

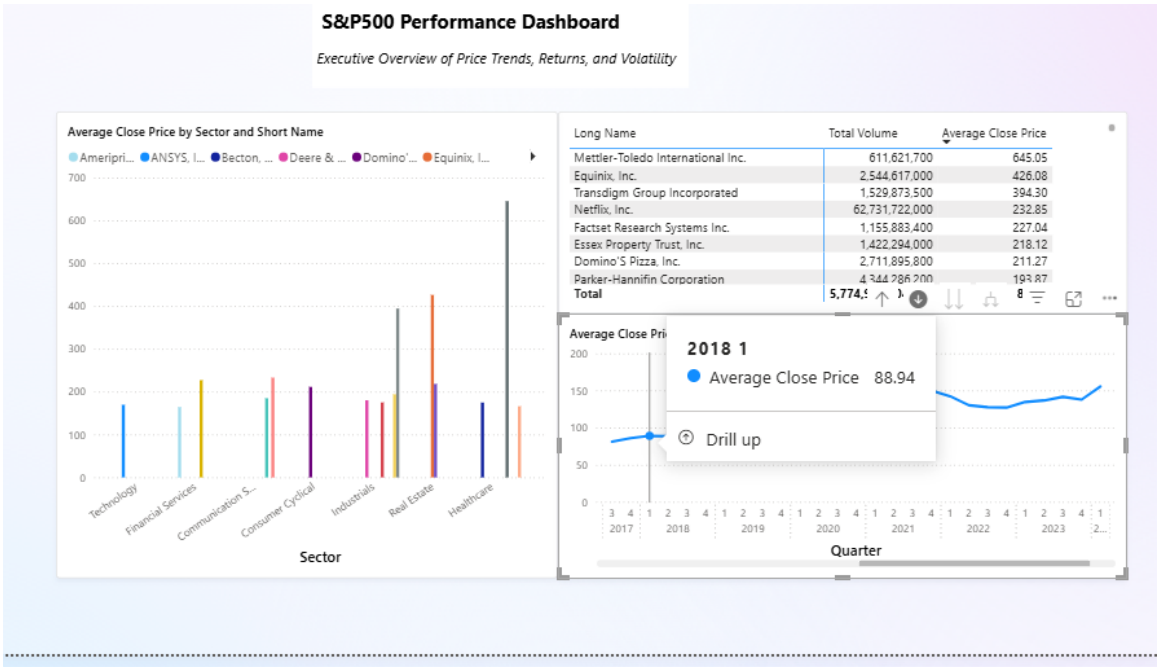


[Figure 13: Executive Summary Dashboard]

Page 2: Detailed Analysis

This page provides deeper operational analysis using matrix analysis, drill-down trend analysis, company-level breakdown, and sector and country comparison.

This supports detailed performance evaluation.

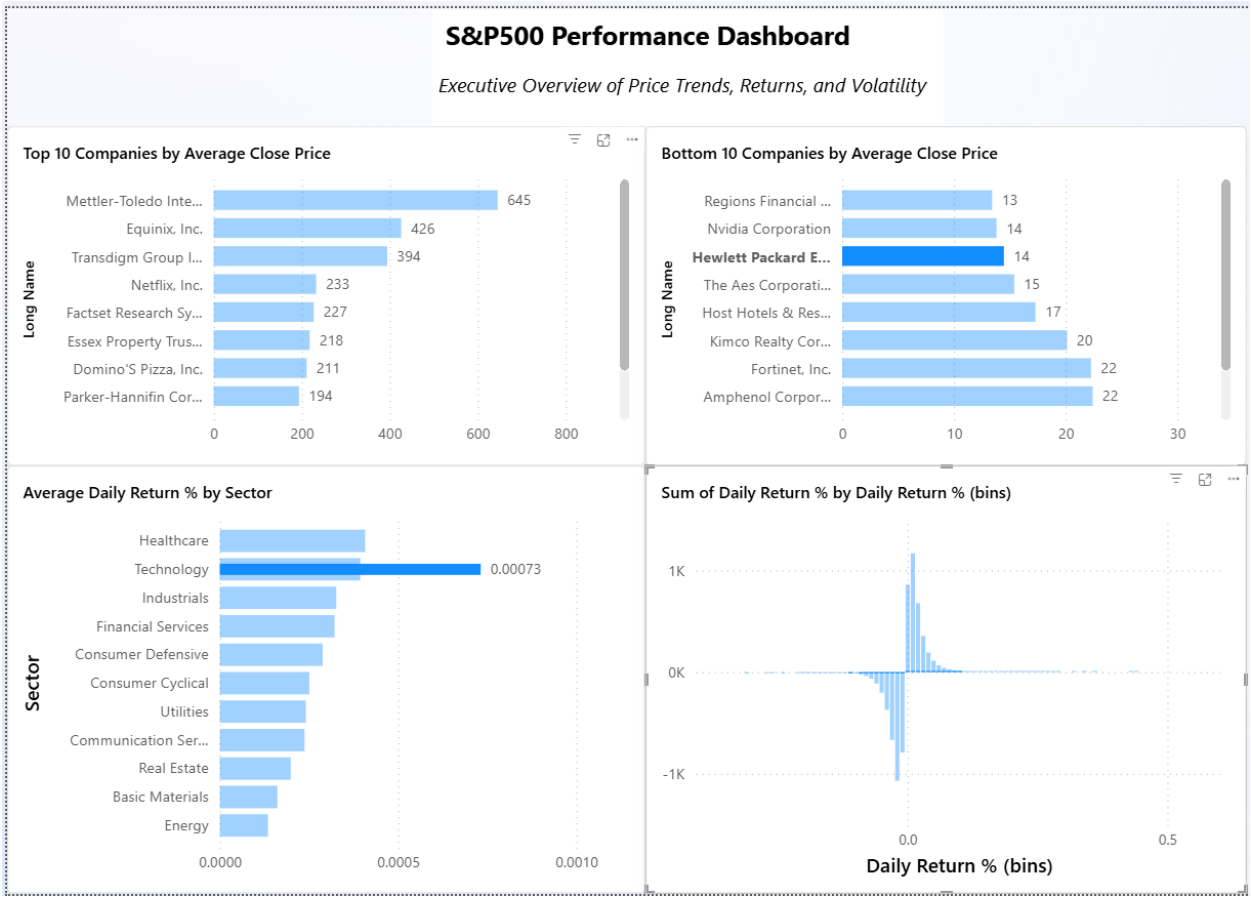


[Figure 14: Detailed Analysis Dashboard]

Page 3: Insights and Performance

This page focuses on advanced performance analysis using Top 10 Performing Companies, Bottom 10 Performing Companies, Sector Growth Comparison, Volatility Distribution, and cross-filter interaction proof.

This supports strategic investment insights.



[Figure 15: Insights and Performance Dashboard]

SECTION 7: ANALYTICAL INSIGHTS AND RECOMMENDATIONS

Analytical Insights

1. Technology and growth-oriented sectors consistently outperformed.
   Sector-level analysis showed that technology-related companies maintained stronger average close prices and higher average daily returns compared to traditional sectors. This indicates stronger investor confidence and sustained long-term growth potential.

2. Market recovery was strong after 2020.
   Trend analysis showed strong market recovery after the 2020 disruption, with a clear upward movement in Average Close Price across major sectors.

3. Volatility is concentrated around small daily return ranges.
   Most trading activity occurred within small daily return ranges, showing that normal market fluctuations are more common than extreme price swings.

4. A small number of companies dominate market performance.
   The Top 10 Companies analysis showed strong market concentration among a few large-cap firms that heavily influence index performance.

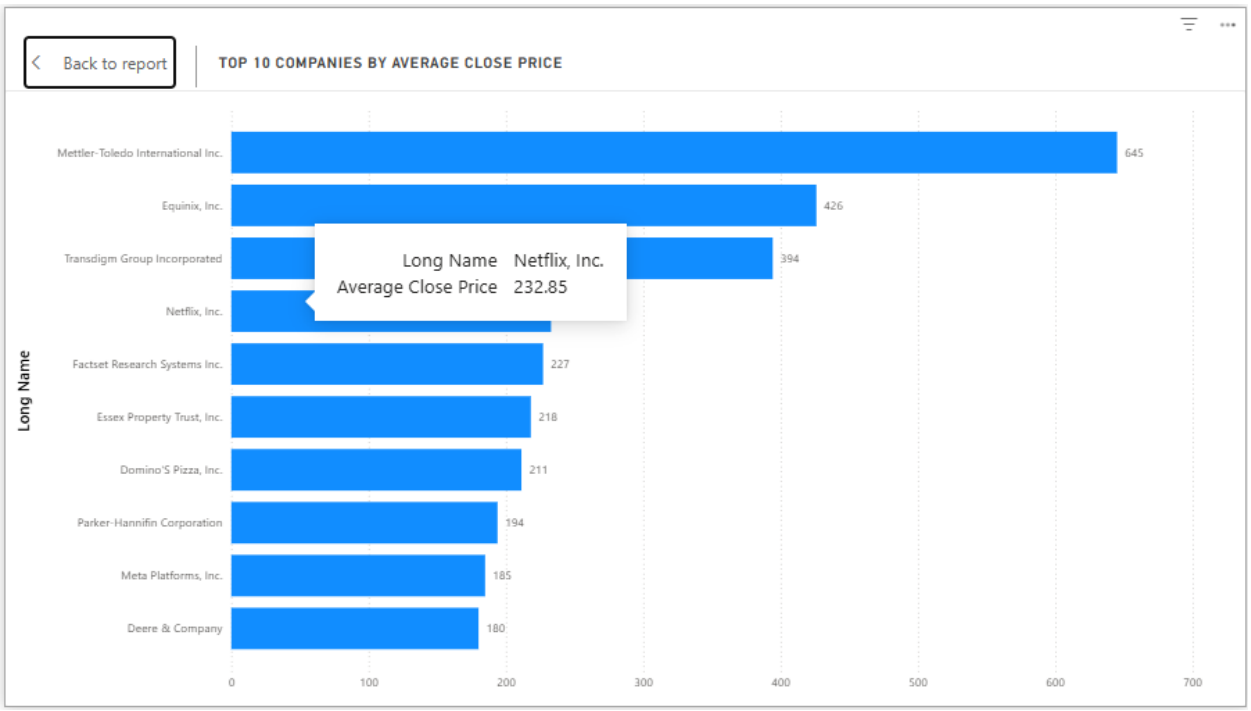
5. Some sectors show persistent underperformance.
   Bottom performer analysis revealed sectors with consistently weak returns and lower average close prices, suggesting long-term operational or structural weakness.

Business Recommendations

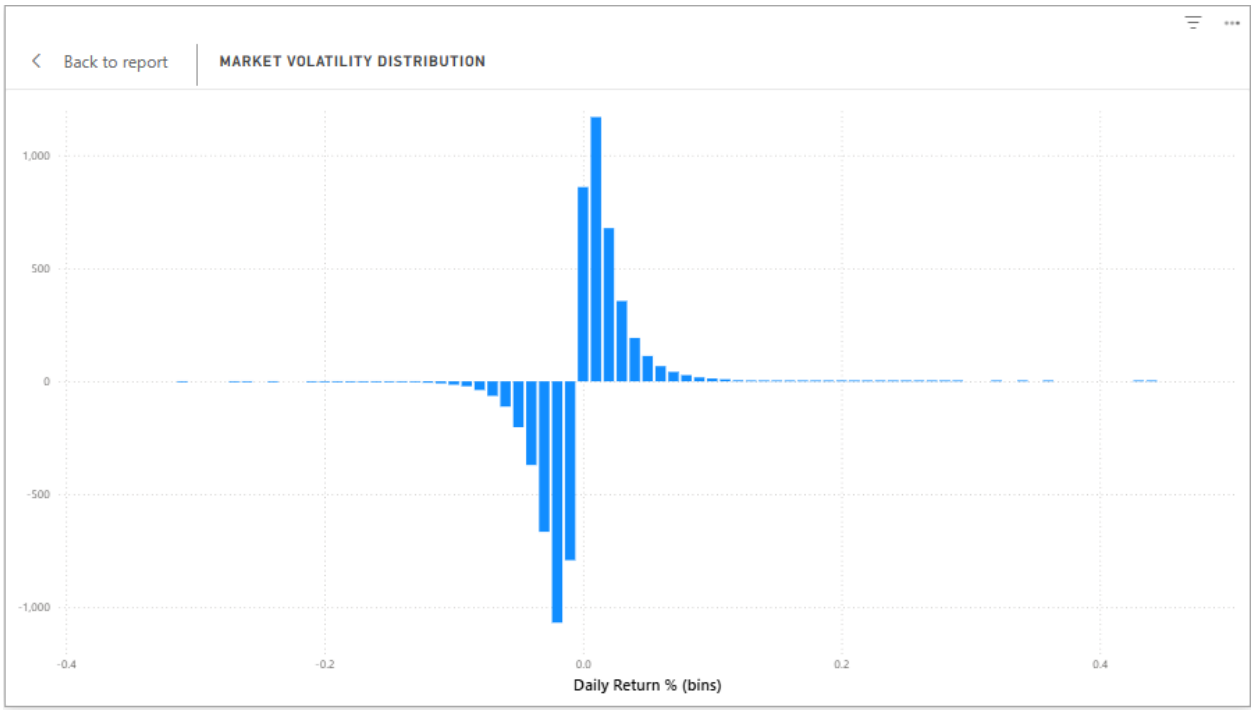
1. Prioritize strong performing sectors.
   Investors should prioritize high-performing sectors such as technology for long-term capital allocation.

2. Monitor volatility for risk control.
   Volatility spikes should be monitored using stop-loss strategies and diversification methods to improve downside protection.

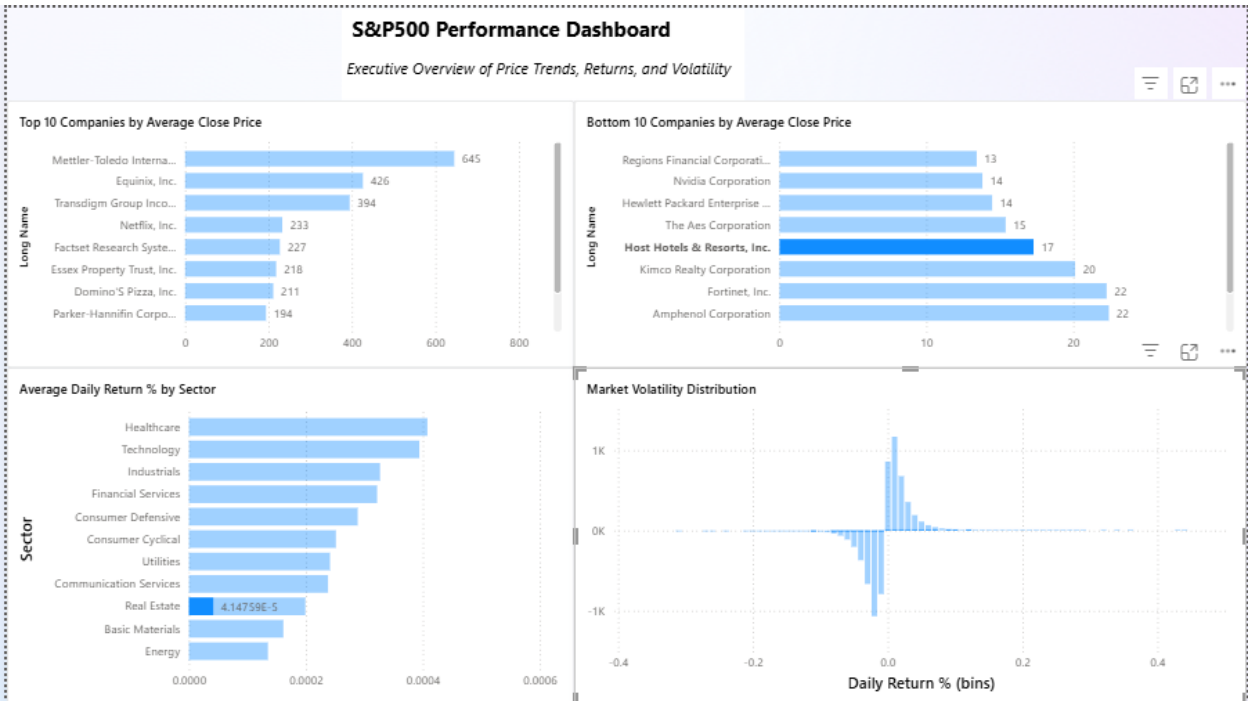
3. Reduce overdependence on large-cap companies.
   Portfolio diversification across multiple sectors should be encouraged to reduce concentration risk.



[Figure 16: Top 10 Companies Analysis]



[Figure 17: Market Volatility Distribution]



[Figure 18: Cross-filtering interaction example]

SECTION 8: GITHUB REPOSITORY AND DEPLOYMENT

A public GitHub repository was created to ensure reproducibility, documentation, and professional project organization.

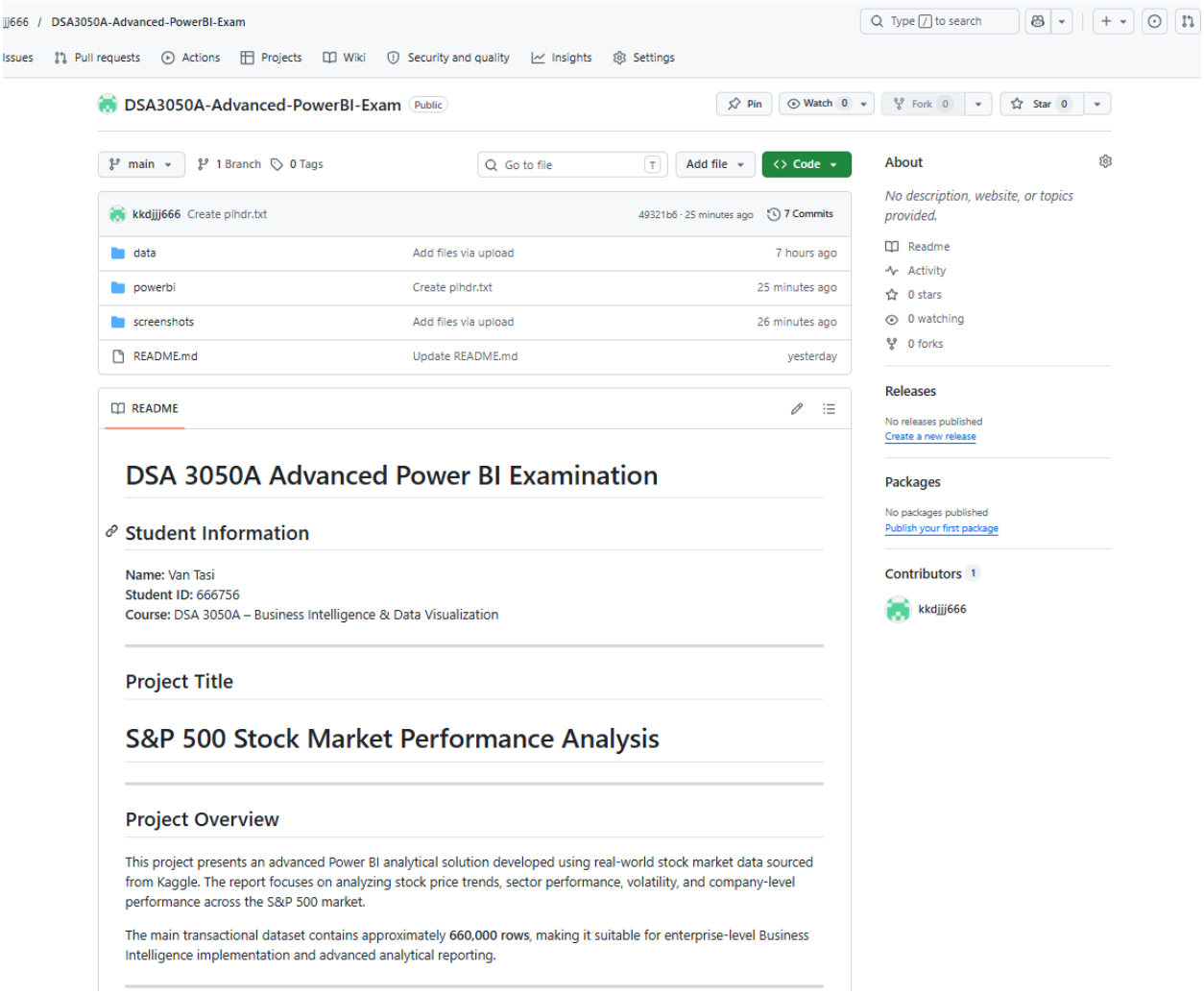
The repository contains datasets, screenshots, the PDF report, Power BI files, and a detailed README.

Due to GitHub upload size limitations, the full transactional dataset of approximately 660,000 rows exceeded direct upload limits. A representative sample file, stocks_sample.csv, was uploaded, while the full dataset was used during Power BI development.

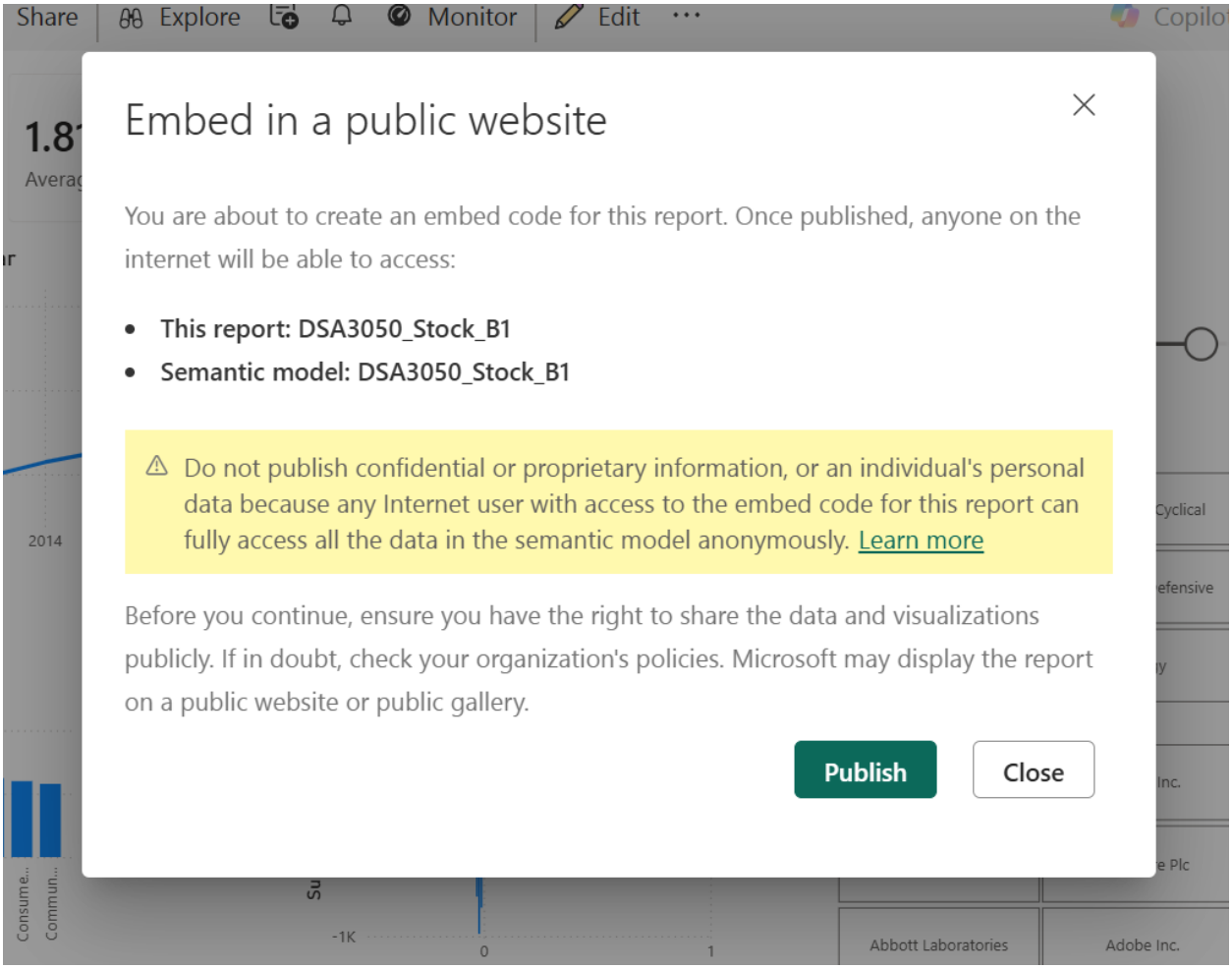
This reflects professional documentation practice.

GitHub Repository:
https://github.com/kkdjjj666/DSA3050A-Advanced-PowerBI-Exam

Power BI Public Dashboard Link:
https://app.powerbi.com/view?r=eyJrIjoiZDY2NmQ4MDctZTM1Yy00NmU1LWE0NWYtZGU2YmI3ZjNhMGQ2IiwidCI6IjE2ZDgzZWU2LTI1NGEtNDY5ZC1hNmNjLTU0ZTJjYTIzMTNlNyIsImMiOjh9



[Figure 19: GitHub repository homepage]



[Figure 20: Power BI publish to web confirmation]

SECTION 9: CONCLUSION

This project successfully demonstrates the complete Business Intelligence lifecycle using Power BI, from raw data preparation to advanced analytical reporting and deployment.

The final solution provides strong decision-support capabilities for stock market performance monitoring and strategic investment analysis.

Through proper modelling, DAX implementation, dashboard design, and deployment, the project reflects practical enterprise-level Business Intelligence implementation within the Banking and Finance domain.

The final dashboard supports both executive decision-making and detailed analytical exploration, making it suitable for real-world business reporting environments.